

Image Plagiarism Detection with Equivariant Encoders

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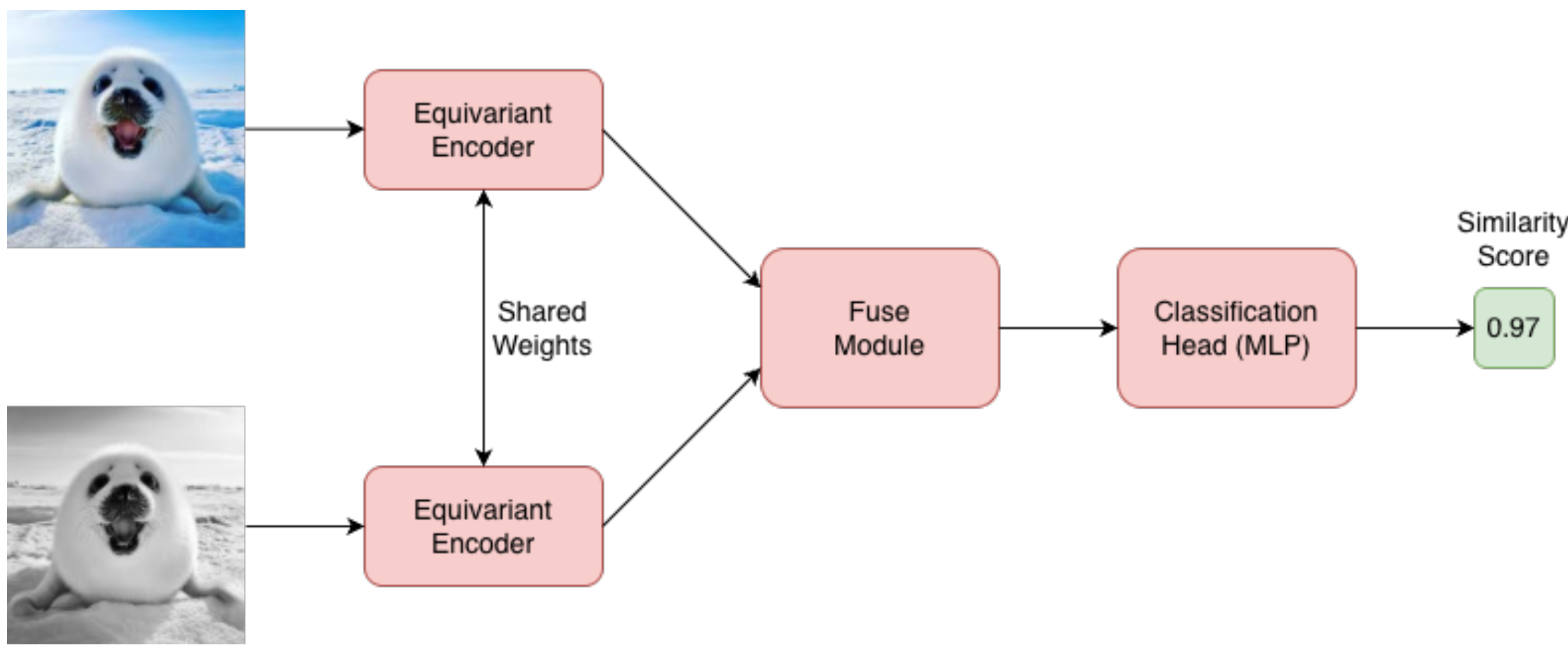
Intelligent Systems, MIPT

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Why Equivariant Encoders

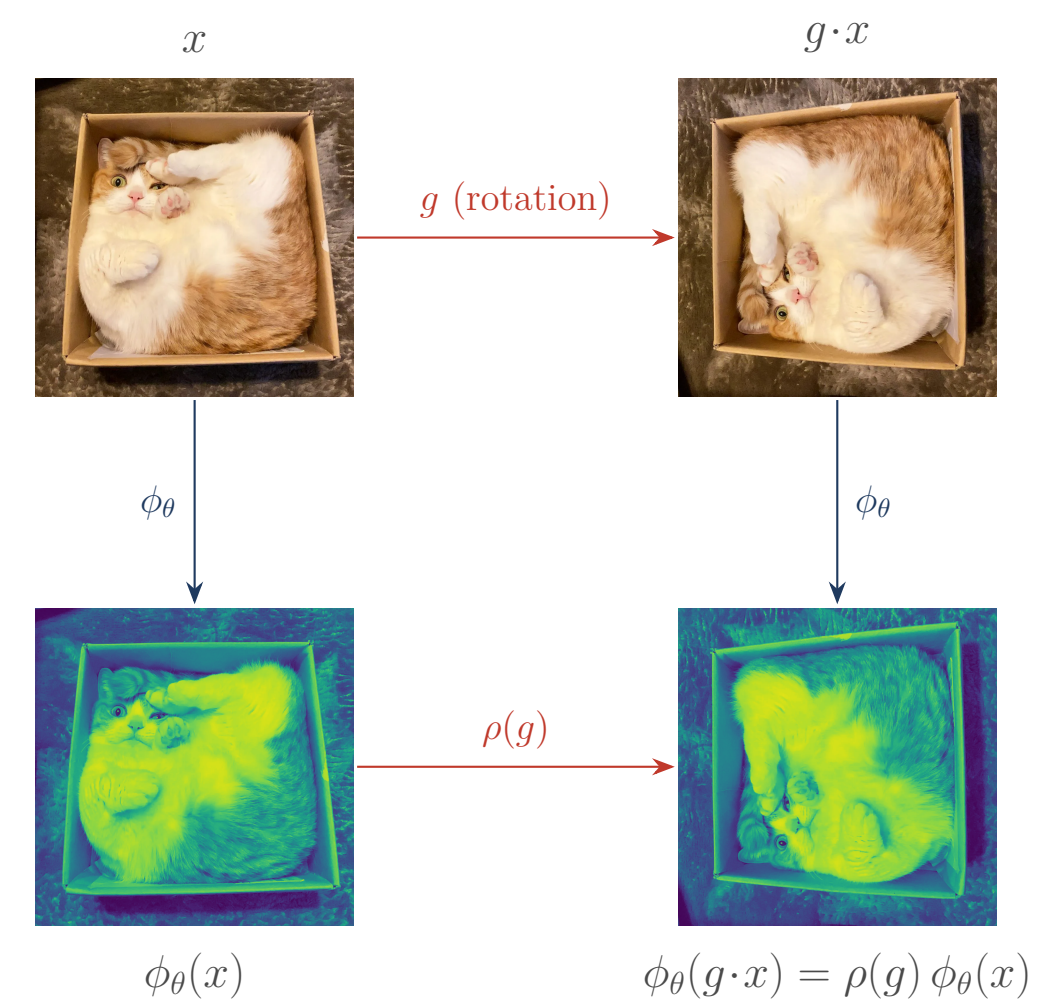
A pirated image is rarely a pixel-perfect copy: rotations, reflections, crops, and JPEG re-compression are routine, and a detector should map all of them to the same point in feature space. Should this invariance be built into the encoder of a siamese plagiarism detector, or approximated through geometric augmentation at training time? To answer this question, we test D_4 -, and $SE(2)$ -equivariant encoders in the task of detecting image plagiarism. Here D_4 is the group of symmetries of the square and $SE(2)$ is the group of continuous roto-translations of the plane.

Siamese Network with a G -invariant Encoder



The detector is $f_\theta(x_1, x_2) = \sigma(h_\psi(\phi_\theta(x_1), \phi_\theta(x_2)))$. The shared encoder $\phi_\theta : \mathcal{X} \rightarrow \mathbb{R}^d$ is chosen G -invariant, so $\phi_\theta(g \cdot x) = \phi_\theta(x)$ for all $g \in G$: geometric variants of one image collapse to a single point. The symmetric fusion head builds $\delta = |z_1 - z_2|$, $p = z_1 \odot z_2$, sets $v = [\delta; p] \in \mathbb{R}^{2d}$, and returns the plagiarism logit through a two-layer MLP.

Equivariance



G -equivariance: the diagram commutes for every $g \in G$. Encoding after a transformation equals transporting the embedding by the representation $\rho(g)$. With $\rho \equiv \text{Id}$ the encoder is G -invariant: $\phi_\theta(g \cdot x) = \phi_\theta(x)$ — geometric variants of x collapse to a single embedding.

Three Encoders, One Pipeline

ViT-Tiny/16 — baseline

224×224 input split into 14×14 patches of 16×16, each linearly projected to $\mathbb{R}^{192}$; twelve transformer blocks, [CLS] embedding as image representation $z \in \mathbb{R}^{192}$. No equivariance guarantee.

Octic ViT — D_4 -equivariant

Each transformer layer replaced by a D_4 -equivariant variant. Octic ViTs yield more computationally efficient networks while also improving performance.

Harmformer — $SE(2)$ -equivariant

Patch embeddings replaced by circular harmonic filters acting as steerable feature maps, combined with self-attention to give continuous roto-translational equivariance over $SE(2)$.

Loss Function

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \lambda \mathcal{L}_{\text{contr}}$$

Class-weighted BCE on the detector probability $\hat{y} = f_\theta(x_1, x_2)$, with $w_+/w_- = \sqrt{B-1}$ to counter the 1:($B-1$) imbalance of the cross-batch pairing protocol.

L^2 **contrastive regularizer** on embeddings $z_i = \phi_\theta(x_i)$ to push embeddings further apart along the L^2 norm in the latent space. :

$$\mathcal{L}_{\text{contr}}(x_1, x_2, y) = \begin{cases} \|z_1 - z_2\|_2^2, & \text{if } y = 1, \\ \max(0, m - \|z_1 - z_2\|_2)^2, & \text{if } y = 0, \end{cases} \quad (1)$$

Experimental Protocol

Training. COCO 2017. Pairs follow the cross-batch protocol: a clean anchor branch + an augmented branch, B^2 ordered pairs per mini-batch with $y_{ij} = \mathbb{1}[i = j]$, label ratio 1:($B-1$).

Test. DomainNet, six domains (real, painting, clipart, quickdraw, infograph, sketch), 200 source images each, 1200 total. *Overall regime:* one 1200-pair balanced test, training-time augmentation on both branches. *Per-transformation regime:* one 1200-pair balanced test per geometric T , photometric off, only T on the positive view.

Augmentation blocks (each transform independent, $p = 0.3$). *Photometric (all runs):* CJ, Gaussian blur, Gaussian noise, JPEG, watermark. *D_4 block (only on ViT-Tiny baseline):* R_{90^k} , $k \in \{1, 2, 3\}$, HFlip, VFlip. *$SE(2)$ block (only on ViT-Tiny baseline):* continuous rotation $\alpha \sim \text{Unif}[0, 2\pi)$, isotropic translation up to $\pm 20\%$.

Geometric Transformations Probed

Name	g	Probes
R_{90}	90° rotation	D_4
R_{180}	180° rotation	D_4
R_{270}	270° rotation	D_4
HFlip	horizontal flip	D_4 reflection
VFlip	vertical flip	D_4 reflection
R_{45}	45° rotation	$SE(2) \setminus D_4$
R_{135}	135° rotation	$SE(2) \setminus D_4$
$T_{0.2}$	translation $\pm 20\%$	$SE(2)$ translation

Results on DomainNet

Model	Overall		Per-transformation Recall (%)							
	FPR %	Recall %	R_{90}	R_{180}	R_{270}	HFlip	VFlip	R_{45}	R_{135}	$T_{0.2}$
ViT-Tiny/16 (photometric only)	7.67	99.83	11.67	16.50	10.67	31.33	21.17	0.50	0.33	1.17
ViT-Tiny/16 + D_4 augs	3.67	100.00	100.00	100.00	100.00	100.00	100.00	0.33	0.33	1.00
Octic ViT	1.17	99.83	100.00	100.00	100.00	100.00	100.00	0.33	0.33	0.50
ViT-Tiny/16 + $SE(2)$ augs	16.17	95.83	100.00	100.00	100.00	100.00	100.00	31.00	30.67	99.17
Harmformer	12.00	99.83	100.00	100.00	100.00	94.17	94.17	0.67	0.67	2.50

Overall metrics use each model's training-time augmentation on both branches; per-transformation Recall isolates one geometric transformation at a time with photometric off. Best overall FPR in bold.

Discussion and Conclusion

D_4 — **equivariance wins.** Both ViT-Tiny+ D_4 and Octic ViT reach 100% Recall on every D_4 transformation, so architectural D_4 -invariance fully replaces the augmentation block. At matched Recall, Octic ViT lowers overall FPR from 3.67% to 1.17%, more than tripling precision at $\tau = 0.5$. Both still fail on off-orbit R_{45} , R_{135} , $T_{0.2}$ ($\leq 1\%$), as expected for invariance restricted to D_4 .

$SE(2)$ — **mixed.** ViT-Tiny+ $SE(2)$ reaches 100% on the D_4 orbit and 99.17% on $T_{0.2}$ but only $\sim 31\%$ on R_{45} / R_{135} , with the highest overall FPR (16.17%). Harmformer reaches 100% on D_4 rotations and 94.17% on the two flips, but $\leq 0.67\%$ on R_{45} / R_{135} and 2.50% on $T_{0.2}$.

Why $SE(2)$ fails — input resolution, not the principle. Bilinear interpolation at off-axis angles introduces enough discretization noise to break analytic $SE(2)$ -equivariance. The D_4 orbit is unaffected because right-angle rotations act by axis-aligned pixel permutations. The augmented ViT-Tiny shows the symmetric failure.

Conclusion. For D_4 the equivariance hypothesis is supported and the augmentation block can be dropped at no loss. For $SE(2)$ the binding constraint is input resolution rather than the equivariance principle.

Future work. rerun at higher resolution; broaden transformations to scale and projective perturbations.